Probability

MATH 110: General Education Math

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Probability

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Sample Spaces and Events

- Define experiment: an action whose result is uncertain.
- Define outcome: one possible result. Define sample space S: the set of all outcomes.
- Toss a coin: $S = \{\mathrm{H}, \mathrm{T}\}$. Roll a die: $S = \{1, 2, 3, 4, 5, 6\}$.
- Clinical: a screening test comes back $S = \{\text{positive}, \text{negative}\}$.

Sample Spaces and Events

- Experiments in several steps: toss a coin twice, $S = \{\mathrm{HH}, \mathrm{HT}, \mathrm{TH}, \mathrm{TT}\}$.
- Toss a coin three times: tree diagram, 8 outcomes, by the Fundamental Principle of Counting (C1): $2 \cdot 2 \cdot 2 = 8$.
- Order matters in an outcome: HT and TH are different outcomes.

Sample Spaces and Events

- Define event: a set of outcomes (a subset of S). The event happens if the outcome is in it.
- Die: "an even number" is $\{2, 4, 6\}$.
- Two tosses: "at least one head" is $\{\text{HH}, \text{HT}, \text{TH}\}$.
- Three tosses: "exactly two heads" is $\{\text{HHT}, \text{HTH}, \text{THH}\}$.

Probability by Counting

- Define probability of an event, $P(E)$: a number from 0 to 1 measuring how likely E is. 0 means impossible, 1 means certain.
- Three ways to write it: $\frac{1}{4} = 0.25 = 25\%$. Practice converting (arithmetic warm-up).

Probability by Counting

- Define equally likely outcomes: outcomes that all have the same chance (fair coin, fair die).
- When the outcomes are equally likely: $P(E) = \dfrac{|E|}{|S|}$ (cardinality, from Sets).
- Die, even number: $\frac{3}{6} = \frac{1}{2} = 0.5 = 50\%$.
- Two tosses, at least one head: $\frac{3}{4} = 0.75 = 75\%$.

Probability by Counting

- Three tosses: list all 8 outcomes and the number of heads in each.
- $P(0 \text{ heads}) = \frac{1}{8}$, $P(1 \text{ head}) = \frac{3}{8}$, $P(2 \text{ heads}) = \frac{3}{8}$, $P(3 \text{ heads}) = \frac{1}{8}$.
- These add to 1. They always do: every outcome is counted exactly once.

Probability by Counting

- Caution: "0, 1, 2, or 3 heads" are four possibilities, but they are not equally likely. $P(1 \text{ head}) \neq \frac{1}{4}$.
- Count the outcomes in the sample space (sequences of tosses), not the possible answers.
- This unequal pattern (1, 3, 3, 1) is the first glimpse of the shape we see in Statistics.

Mutually Exclusive Events and the Addition Rule

- Define mutually exclusive events: events with no outcomes in common, i.e., disjoint events, $A \cap B = \varnothing$ (Sets). They cannot both happen.
- Die: "1 or 2" and "5 or 6" are mutually exclusive. "Even" and "greater than 3" are not (both contain 4 and 6).

Mutually Exclusive Events and the Addition Rule

- Define the addition rule: if A and B are mutually exclusive, $P(A \cup B) = P(A) + P(B)$.
- Why: no outcome is counted twice.
- Three tosses: $P(\text{at least 2 heads}) = P(2 \text{ heads}) + P(3 \text{ heads}) = \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$.

Mutually Exclusive Events and the Addition Rule

- Recall the complement from Sets. For an event, A^c is all outcomes in S not in A.
- A and A^c are mutually exclusive and together make up S, so $P(A) + P(A^c) = 1$ and $P(A^c) = 1 - P(A)$.
- Three tosses: $P(\text{at least one head}) = 1 - P(\text{no heads}) = 1 - \frac{1}{8} = \frac{7}{8}$.

Mutually Exclusive Events and the Addition Rule

- Subtracting: "at most 2 heads" splits into "0 heads" and "1 or 2 heads" (mutually exclusive).
- So $P(\text{1 or 2 heads}) = P(\text{at most 2 heads}) - P(\text{0 heads}) = \frac{7}{8} - \frac{1}{8} = \frac{3}{4}$.
- This is exactly how we find the area between two values with a z-table.

Independence and the Multiplication Rule

- Define independent events: events where knowing that one happened does not change the probability of the other.
- Independent: two tosses of a coin (the coin has no memory); the results of two unrelated patients.
- Not independent: a ward has 10 patients, 3 with the flu. Pick one patient, then a second from the remaining 9. Whether the second has the flu depends on the first ($\frac{3}{9}$ or $\frac{2}{9}$).

Independence and the Multiplication Rule

- Define the multiplication rule: if A and B are independent, $P(A \cap B) = P(A) \cdot P(B)$.
- Check against counting: two tosses, $P(\text{HH}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$, the same as $\frac{1}{4}$ from the sample space.
- Toss a coin and roll a die. With $H =$ "heads" and $X =$ "rolls a 6": $P(H \cap X) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$ (check with the 12-outcome sample space).

Independence and the Multiplication Rule

- Any particular sequence of n tosses: $P(\text{HHT}) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$; in general $\left(\frac{1}{2}\right)^n$. This is why all 2^n sequences are equally likely.
- The rule also works when outcomes are not equally likely, where counting fails: a treatment works for 80% of patients. For two patients, $P(\text{both respond}) = 0.8 \cdot 0.8 = 0.64$ and $P(\text{neither responds}) = 0.2 \cdot 0.2 = 0.04$.
- $P(\text{at least one responds}) = 1 - 0.04 = 0.96$ (complement).

Conditional Probability

- Define two-way table: a table counting a group sorted by two characteristics at once.
- Example: 1000 people screened for a condition. 100 have it: 90 test positive, 10 test negative. 900 do not: 45 test positive, 855 test negative.
- Table with rows (has condition, does not), columns (positive, negative), and totals: 135 positive, 865 negative.
- Read probabilities straight from the table: $P(\text{positive}) = \frac{135}{1000} = 0.135$.

Conditional Probability

- Define conditional probability $P(A \mid B)$: the probability of A, given that B happened.
- Given B, only the row or column for B matters. $P(\text{positive} \mid \text{has condition}) = \frac{90}{100} = 0.90$.
- $P(\text{has condition} \mid \text{positive}) = \frac{90}{135} \approx 0.67$.
- These are not the same. A positive result means only a 2-in-3 chance of having the condition, even though the test catches 90% of cases.

Conditional Probability

- Formula: $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$.
- Check: $P(\text{has condition} \mid \text{positive}) = \frac{0.090}{0.135} \approx 0.67$, as from the table.
- Why the gap: the 45 false positives come from the much larger group of 900 people without the condition.

Conditional Probability

- Independence, revisited: A and B are independent exactly when $P(A \mid B) = P(A)$.
- Screening table: $P(\text{positive}) = 0.135$ but $P(\text{positive} \mid \text{has condition}) = 0.90$. Not independent, which is what we want from a test.
- Ward example from Independence: $P(\text{second has flu} \mid \text{first has flu}) = \frac{2}{9} \neq \frac{3}{10}$.

Binomial Experiments

- Define binomial experiment: the same two-outcome trial (success or failure) repeated a fixed number of times, with independent trials and the same probability of success p on each; we count the number of successes.
- Examples: toss a coin 10 times and count heads ($p = \frac{1}{2}$); treat 4 patients with a treatment that works 80% of the time and count responses ($p = 0.8$).

Binomial Experiments

- Four tosses: $2^4 = 16$ sequences (C1), each with probability $\left(\frac{1}{2}\right)^4 = \frac{1}{16}$ (multiplication rule).
- Sequences with exactly 2 heads: choose which 2 of the 4 tosses are heads, ${}_4C_2 = 6$ (C2). List them: HHTT, HTHT, HTTH, THHT, THTH, TTHH.
- These are mutually exclusive, so $P(\text{exactly 2 heads}) = 6 \cdot \frac{1}{16} = \frac{6}{16}$ (addition rule).

Binomial Experiments

- Same reasoning with $p = 0.8$: exactly 3 of 4 patients respond.
- One such sequence, RRRN: $0.8 \cdot 0.8 \cdot 0.8 \cdot 0.2$. Every sequence with 3 R's and 1 N has this same probability.
- There are ${}_4C_3 = 4$ of them, so $P = 4 \cdot (0.8)^3(0.2) \approx 0.41$.
- In general: $P(\text{exactly } k \text{ successes}) = {}_nC_k\, p^k(1-p)^{n-k}$.

Binomial Experiments

- For a fair coin, $p = \frac{1}{2}$ and the formula becomes $P(\text{exactly } k \text{ heads}) = \frac{{}_nC_k}{2^n}$.
- Table for 4 tosses: $\frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16}$ (used in Statistics, Experimental Probability).
- The probabilities add to 1.

Binomial Experiments

- Ten tosses, 1024 sequences. $P(5 \text{ heads}) = \frac{{}_{10}C_5}{1024} = \frac{252}{1024} \approx 0.246$.
- $P(\text{at least 9 heads}) = \frac{10+1}{1024} \approx 0.011$ (addition rule).
- $P(\text{at most 8 heads}) = 1 - 0.011 \approx 0.989$ (complement).
- Even the most likely number of heads happens less than a quarter of the time.